

Retail Spatial Intelligence

CCTV Foot-Traffic & Spatial Behavior Report

VERIFIED COMPUTER VISION DATA

Analysis Parameters & Metadata

VIDEO SOURCE:	sample_retail_cctv.mp4	DATE PROCESSED:	Oct 6, 2026, 09:36 PM
RESOLUTION:	768x432 (768x432)	VIDEO DURATION:	49.67 seconds (596 frames @ 12 fps)
ZONE CONFIG:	Custom Drawn Polygons	REPORT VERSION:	v2.1 (Canonical Analytics v2.0)
TRACKING QUALITY:	Good (High Confidence)	ANALYSIS ID:	3c3035e7-a914-4999

UNIQUE VISITORS

7

Valid tracked customers

AVG ZONE DWELL

2.03s

Median: 2s

PEAK OCCUPANCY

3

Concurrent shoppers

TOTAL ZONE VISITS

6

Across 2 zones

Executive Summary

DETERMINISTIC SPATIAL RULES SUMMARY

Video analysis observed 7 unique tracked customer visits across 49.67s of footage. Overall average dwell time was 2.03s with a peak simultaneous store occupancy of 3 people. 1 potential bottleneck area(s) and 0 potential dead-zone area(s) were deterministically identified.

1. Computer Vision Processing & Traffic Diagnostics

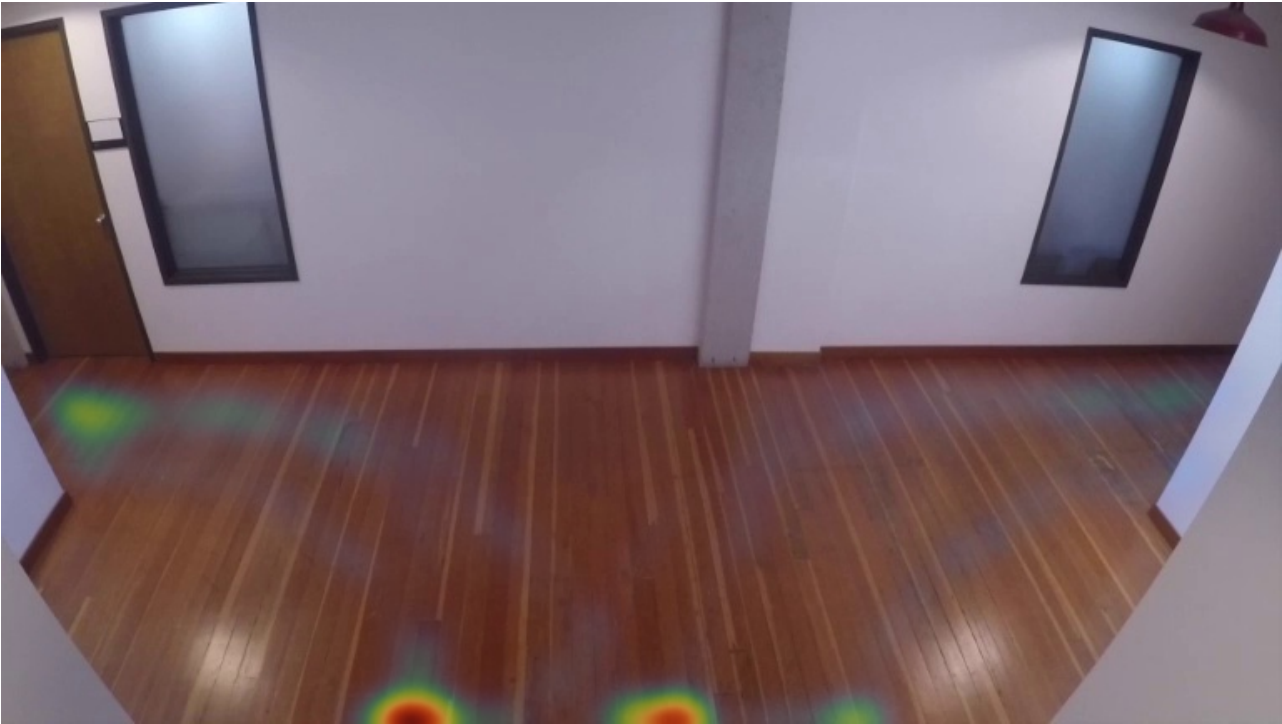
Footage was processed using Ultralytics YOLOv8 for human detection and ByteTrack for continuous anonymous multi-object tracking. Detection ground points were extracted at the bottom-center of bounding boxes ($x = [x1+x2]/2$, $y = y2$) to represent true floor contact. Tracks below the duration threshold of 1s were filtered to remove transient occlusion artifacts.

PROCESSING TELEMETRY

Raw Tracks Detected: 7	Effective Tracking Confidence: Good	Avg Detection Confidence: 0.83
Filtered Transient Tracks: 0	Average Track Duration: 4.5s	Relative Speed: 99 px/s

2. Observed Foot-Traffic Density Heatmap

Heatmap generated via 2D Gaussian Kernel density accumulation of verified customer foot-point coordinates overlaid onto the representative video frame.



Heatmap Methodology & Legend:

Blue = Low Pedestrian Activity | Green/Yellow = Moderate Dwell & Flow | Red/Crimson = Peak Traffic & Congestion Stagnation

3. Spatial Zone Performance Analysis

Deterministic measurements per defined spatial zone. Dwell sessions include a 0.8s occlusion buffer to prevent transient drops from breaking sessions.

Zone Name	Visits	Unique	Avg Dwell	Max Dwell	Peak Occ	Avg Occ	Share (%)	Density
North Corridor Entrance	3	3	1s	1.5s	1	0.05	50%	4.29
South Merchandising Display	3	3	3.05s	3.83s	2	0.19	50%	4

4. Customer Flow & Traversal Pathways

Zone-to-zone transitions mined from consecutive spatial crossings across verified customer paths:

No multi-zone transitions observed during the sample period.

Frequent Customer Journey Sequences:

1. South Merchandising Display

Observed in 3 customer(s) (50% of journeys) - Avg Duration: 4.6s
2. North Corridor Entrance

Observed in 3 customer(s) (50% of journeys) - Avg Duration: 4.4s

5. Forensic Bottleneck & Dead-Zone Intelligence

POTENTIAL BOTTLENECK ANALYSIS:

- **South Merchandising Display (Bottleneck Score: 79/100 - Confidence: Medium)**
 - Evidence: Elevated peak concurrency: 2 simultaneous customer tracks observed
 - Evidence: Prolonged dwell duration: 3.05s exceeds store average (2.03s)
 - Evidence: High spatial traffic density: 4.0 visitors per 100k sq px

POTENTIAL DEAD-ZONE ANALYSIS:

- Healthy foot-traffic distribution observed across all monitored zones.

HIGH-ACTIVITY DESTINATION ZONES:

- South Merchandising Display: 3 visits, 3.05s avg dwell, peak concurrent: 2 (High Activity Corridor)
- North Corridor Entrance: 3 visits, 1s avg dwell, peak concurrent: 1 (High Activity Corridor)

6. Strategic Recommendations & Priority Action Plan

[Congestion Alleviation] South Merchandising Display

Evidence: Elevated peak concurrency: 2 simultaneous customer tracks observed; Prolonged dwell duration: 3.05s exceeds store average (2.03s); High spatial traffic density: 4.0 visitors per 100k sq px

Action: Widen primary aisle clearance or reposition high-interaction displays in South Merchandising Display to alleviate observed transit bottlenecks.

Prioritized Action Plan:

- [High Priority]:** Audit aisle clearance in South Merchandising Display to eliminate movement slowdowns.
- [Medium Priority]:** Compare peak occupancy periods with staff scheduling.
- [Low Priority]:** Conduct subsequent video analysis after implementing spatial adjustments to evaluate before/after metrics.

7. Scientific Methodology & Audit Traceability

- YOLOv8 Person Detection**
Pretrained neural model detecting COCO Class 0 (person). Frame sampling optimizes CPU throughput without altering physical timeline.
- ByteTrack Multi-Object Tracking**
Associates detections using high/low confidence thresholds and Kalman state estimation with persistent anonymous IDs (Customer #1, #2).
- Floor Foot-Point Extraction**
Computes ground contact coordinate at $x = (x1+x2)/2$, $y = y2$ to bypass perspective bounding box tilt.
- Point-in-Polygon Zone Assignment**
Assigns floor foot-points to arbitrary user-drawn polygons or balanced camera regions using OpenCV pointPolygonTest.
- Dwell Time with Occlusion Grace Buffer**
Maintains an 0.8s tolerance buffer before closing dwell sessions to prevent brief tracking drops from artificially splitting sessions.
- 2D Gaussian Density Heatmaps**
Accumulates foot-points into a spatial matrix smoothed with a normalized Gaussian kernel and mapped through OpenCV Turbo colormap.

8. System Limitations & Forensic Transparency

- Single-Camera Monocular Perspective: Field of view is restricted to the camera mounting angle. Speeds are reported in image-space pixels/second, not calibrated metric coordinates.
- Visual Occlusion: Dense crowds or structural pillars can temporarily occlude foot points. The platform uses a 0.8s tolerance buffer, but long occlusions may cause track fragmentation.
- Zero PII Policy: The system operates with transient anonymous IDs only. No face recognition, demographic profiling, or biometric storage is performed.
- No POS Integration: Analysis reflects physical pedestrian movement only. Purchase conversions, basket size, and revenue metrics require external POS pairing.

AUTHENTICATION & VERIFICATION STAMP

Generated deterministically by Retail Spatial Intelligence Platform Engine v2.1. All metrics directly traceable to video frame detections. Job ID: 3c3035e7-a914-4999-9fda-6031e0629eea

